

Whether Emotions are “Engaging” or “Disengaging” Depends on Relationship Functions

Mujtaba K. Chughtai*, Maria Gendron, Margaret S. Clark

Department of Psychology, Yale University, New Haven, Connecticut, USA

*Correspondence concerning this article should be addressed to Mujtaba K. Chughtai:

muji.chughtai@yale.edu

Author Accepted Manuscript

This is the author accepted manuscript of the following article:

Chughtai, M. K., Gendron, M., & Clark, M. S. (2025). Whether emotions are engaging or disengaging depends on relationship functions. *Affective Science*, 6(4), 700–713. <https://doi.org/10.1007/s42761-025-00327-8>

Please cite the published version, available at <https://doi.org/10.1007/s42761-025-00327-8>

This version of the article has been accepted for publication, after peer review (when applicable) and is subject to Springer Nature's AM terms of use, but is not the Version of Record and does not reflect post-acceptance improvements, or any corrections.

Abstract

Researchers have categorized certain emotions as socially engaging (interdependence-affirming) and others as disengaging (independence-affirming) and have shown that these emotions pattern differently across cultures differing in dominant self-construals. However, the social functions different relationships serve and the associated nature of interdependence that characterize relationships also differ within cultures. Across four samples, we examined how frequently Americans report experiencing, expressing, and perceiving eight emotions in the context of normatively communal and transactional relationships. We also examined how engaging and functional these emotions were reported to be. In Studies 1 (N = 393) and 2 (N = 394), participants reported experiencing, expressing, and perceiving both putatively engaging (respect, guilt) and disengaging (pride, anger) emotions more frequently in communal relationships, and each emotion was reported as more engaging in communal than in transactional relationships. In Studies 3 (N = 353) and 4 (N = 330), we examined the patterning of additional emotions that past work indicates are particularly tied to communal relationships (empathic joy, empathic sadness, hurt, love) across the same types of relationships as examined in Studies 1 and 2. All these emotions were more likely to be perceived and expressed in communal (versus transactional) relationships, and they were all judged to be more engaging and functional in communal than in transactional relationships. Additionally, participants believed expressing these emotions within transactional relationships would be disconnecting, relationship-undermining, and distancing (i.e., disengaging). In sum, whether emotions are “engaging” or “disengaging” appears to depend upon on how they fulfill the social functions of distinct relationships.

(Words: 250); *Keywords: relationship types, social engagement dimension, culture*

Scholars have argued for a social orientation dimension of emotions, such that certain emotions are more socially engaging while others are more socially disengaging (e.g., Kitayama et al., 2006). They have identified certain emotions as engaging (e.g., respect and guilt) and others as disengaging (e.g., pride and anger), based on theorizing that engaging emotions promote interdependence and connection while disengaging ones promote independence and disconnection. Supporting these putative categorizations, they have shown that their putatively engaging emotions are more commonly experienced in an interdependent culture (Japan) than an independent one (the United States) and that the reverse holds true for their putatively disengaging emotions. These studies demonstrated that cultural norms for interdependence predictably influence emotional lives.

However, *within* cultures, the nature of interdependence that people have with others also depends on the type of relationship they have with them, as different social functions (Bugental, 2000) and ways of relating and sharing benefits (Fiske, 1992; Rai & Fiske, 2011) are embedded within different types of social relationships (Clark & Mills, 1979; 2012 Earp et al., 2021; 2025). Cooperative relationship functions include giving and receiving care broadly and non-contingently (care), providing for the fair exchange of goods and services on a contingent basis (transaction), providing for hierarchical relationships in which leaders and followers exist for their mutual benefit (hierarchy), and providing for sexual relationships (mating; Earp et al., 2021; 2025). One way to characterize various relationships is by the cooperative functions they do or do not serve. Communal relationships serve a broad mutual care function (Clark & Mills, 1979; 2012). Communal partners assume responsibility for one another's welfare, track one another's needs and desires (Clark et al., 1989), and give and seek non-contingent responsiveness to their own needs (Clark et al., 1987). Transactional relationships are also cooperative relationships, but

they serve different functions than communal ones. Transactional relationships are formed to allow for fair exchanges of goods and services, such that transactional partners track input and outcomes from these relationships (Clark, 1984), working to keep things balanced and fair. The different functions these relationships serve and the interdependence defining these relationships have important implications for people's emotional lives (Clark, et al., 2025).

Past work has shown that the more communal a relationship is, the more individuals within these relationships express (Barret et al., 1998; Von Culin et al., 2018, Ruan et al., 2020, Ruan et al., 2024), and positively react to emotion (Clark & Taraban, 1991). Similarly, Le and Reis (2024) found that participants' perceptions of communal motivation in their relational partners positively predict the likelihood of those participants sharing happiness, pride, gratitude, sadness, anxiety, guilt, and anger toward those partners. So, Americans seem to realize the benefits of expressing emotions (including putatively disengaging emotions) in communal compared to transactional relationships, aligning with prior work that showed emotion expression elicits support in communal but not transactional contexts (Clark et al., 1987). In contrast, Americans are often unwilling and uncomfortable expressing emotions in transactional relationships, especially emotions that facilitate/signal care.

In these studies, the researchers did not directly assess the interpersonal functions or engaging/disengaging nature of expressing these emotions across relational contexts. So, it is possible that the *same* emotions may be differentially engaging or disengaging *across* different relationships within the same broad cultural context. In general, we believe that emotions that facilitate the functions that a relationship serves will be experienced and expressed more often, while also being perceived as more engaging. In contrast, emotions that interfere with the functions that a cooperative relationship serves will be experienced and expressed less often,

while also being perceived as more disengaging. To test the validity of these general assumptions, we focus on emotions in two distinct types of cooperative relationships within the United States: communal and transactional relationships.

The Present Research

In Studies 1 and 2, we examined how frequently Americans report experiencing, expressing, and perceiving two putatively engaging (respect, guilt) and two putatively disengaging (pride, anger) emotions in the context of communal and transactional relationships. To measure the engaging/disengaging nature of these emotions, we also examined how connected participants felt when expressing and perceiving these emotions in these relational contexts. In Study 1 we focused on participants' reports of their own emotions, and in Study 2 we focused on participants' reported perceptions of the same emotions in their partners.

To further test our overall claims regarding emotion experience, expression, and perception, in Studies 3 and 4, we examined four new emotions (empathic joy, empathic sadness, hurt, love) explicitly chosen because they serve the care function. We also added additional measures that more directly test the functions of expressing and perceiving these emotions. The first was whether the emotion draws them close versus distances them from their partner. The second was whether the emotion affirms or undermines a participant's relationship with their partner. Additionally, we did not limit questions regarding the engaging/functional nature of emotions to only participants who have expressed/perceived each emotion in each context. This allowed us to examine whether participants who never express/perceive an emotion in a given context have different beliefs about the interpersonal effects of these emotions than those who have expressed/perceived them. Studies 1 and 3 focused on participants' own expressions, and Studies 2 and 4 were direct mirrors focused on perception of partners' emotions.

Study 1

Method

Hypotheses

Preregistered hypotheses for all studies are on this project's OSF page: <https://osf.io/jcxat/>. We hypothesized that participants would report experiencing the putatively engaging emotions more frequently in communal (versus transactional) relationships (H1). We also hypothesized that, in communal relationships, participants would report experiencing the putatively engaging emotions more frequently than the disengaging ones. In contrast, in transactional relationships, we hypothesized that the reported frequencies of experience would be similarly low for both types of emotions (H2). The hypotheses were the same for the frequency of expression (H3, H4).

The next set of hypotheses pertain to reported feelings of connection when expressing emotions. We hypothesized that participants would report feeling more connected while expressing emotions in communal (versus transactional) relationships (H5). We also hypothesized that participants would report feeling more connected with their relationship partner when expressing the putatively engaging (versus disengaging) emotions (H6). Lastly, we hypothesized that reported feelings of connection when expressing an emotion would be positively related to the reported frequency of expressing that emotion within communal (but not transactional) relationships (H7).

Participants

We conducted an a-priori power analysis using the *sjstats* package in R (Lüdtke, 2024; R Core Team, 2024) to determine the required sample size to detect a small-medium effect size (Cohen's $d = 0.3$) for the interaction between emotion type and relationship type on the reported

frequency of emotion experience. Based on this analysis, we recruited 403 individuals to participate in Study 1. All participants were recruited and prescreened through Prolific, where eligible participants could choose to participate after reading a short description of the study. Individuals were eligible to participate in Study 1 if they were currently married, engaged, in a romantic relationship, or in a civil partnership/civil union.¹ Additionally, to control for potential variation due to language and culture, individuals were eligible to participate in Study 1 if their first language was English, their nationality was the United States, and they were in the United States at the time of the survey.

After removing data from participants who did not finish the survey ($n = 1$) and participants who failed at least one or more attention checks ($n = 9$), we retained responses from 393 unique individuals. 79.1% were White, 7.4% were Asian, 6.4% were Black, 4.6% were multiple races, 2.3% were some other race, and 0.2% did not disclose their race. The mean age of retained participants was 40.3 years ($SD = 12.7$), and 2 participants did not disclose their age. 53.9% of retained participants were female, 45.3% were male, and 0.7% preferred not to report their sex.²

Procedure

Consent to participate and all measures for Study 1 were administered via an online anonymous Qualtrics survey. First, participants read short descriptions about experiencing and expressing emotions. They were then asked to take a moment to think about individuals who fit

¹ We did not require participants to be in school or employed. We expected that all our participants would be able to report on interactions with classmates/colleagues from some point in their lives, even if they were not currently in school or employed.

² These demographics were based on Prolific's demographic data, which is why we report "Sex" instead of "Gender." In future studies, we include our own demographic questionnaire in our surveys and collect and report demographics that include gender instead of sex.

four different lay-language identified relationships and to consider typical interactions with these individuals when responding to later measures. Two of the relationships were normatively communal in nature (romantic partner, close friend); two were normatively transactional in nature (classmate/colleague, stranger).

The participants responded to questions pertaining to sixteen total scenarios, consisting of every combination of the four relationships and four emotions (two putatively engaging and two putatively disengaging). The four emotions were “respect for them,” “guilt toward them,” “pride in yourself,” and “anger at them.” As already mentioned, Kitayama et al. (2006) classified the former two emotions as socially engaging and the latter two as socially disengaging.

First, participants were asked the frequency with which they experience each emotion when they are with their specified relationship partner(s). Next, they were asked the frequency with which they express the given emotion when they are with their specified relationship partner(s). Both frequency questions were rated on the following 5-point scale: “Never,” “Little of the time,” “Some of the time,” “Most of the time,” and “All of the time.” For data analysis, the responses for the frequency questions were coded 0, 1, 2, 3, and 4, respectively. Last, if the participant indicated that they express a given emotion to a given relationship partner at least “a little of the time,” they were asked how connected they feel to their relationship partner(s) when they express that emotion. The connection question was rated on the following scale: “Very disconnected,” “Disconnected,” “Neither connected or disconnected,” “Connected,” and “Very connected.” For data analysis, responses to the connection question were coded -2, -1, 0, 1, and 2, respectively. For example, for the “close friend” and “pride in yourself” scenario, participants were asked “When you are with your close friends, how often do you experience pride in yourself?” then “When you are with your close friends, how often do you express pride in

yourself?,” and lastly, “When you express pride in yourself, how connected do you feel with your close friends?”

All questions for a given relationship were asked in succession. The order of the relationships and the order of the emotions within each relationship block was randomized for all participants. Additionally, embedded within the survey were two attention check questions that resembled the measures described above, but asked the participants to indicate a specific response. These attention check questions were placed randomly within the survey, and participants were retained for data analysis if they responded in accordance with the attention check instructions.

Results

For descriptive statistics, between-subject variability assessments, and model assumption checks for all studies, see the online supplemental materials. For all analyses, we fit ANOVA models and assessed pairwise comparisons after post-hoc Tukey adjustments.³

The interaction between relationship type and emotion type on reported frequency of experience was significant ($\beta = -0.38$, $t(6281) = -6.46$, $p < .001$, 95% CI $[-0.490, -0.262]$). That is, participants reported experiencing putatively engaging emotions more frequently in communal versus transactional relationships (pre-registered as H1). In communal relationships, participants reported experiencing putatively engaging emotions more frequently than disengaging ones, but in transactional relationships, they reported experiencing the emotion

³ To ensure that the categorization of “strangers” as a transactional relationship was not undermining our results, we fit exploratory models for our three variables from Study 1 where we removed “strangers” from the analysis (retaining only “classmates/colleagues” in the transactional relationship category). All effects between relationship types for frequencies of experiencing and expressing each individual emotion, as well as the feelings of connection associated with expressing each emotion, remain significant.

types at similar frequencies (pre-registered as H2). Participants reported experiencing respect, pride, and guilt more frequently in communal than transactional relationships, suggesting even putatively disengaging emotions are more engaging within communal relationship contexts. Further, the difference in reported experience between relationship types was significantly larger for respect than for pride, for pride than for guilt, and for guilt than for anger. Participants reported experiencing anger with a similar frequency in both relationship types. These results are presented in Figure 1, see also Table 1.

The interaction between relationship type and emotion type on reported frequency of expression was also significant ($\beta = -0.12$, $t(6178) = -2.16$, $p = .031$, 95% CI $[-0.236, -0.011]$). As with experience, participants reported expressing putatively engaging emotions more frequently in communal than transactional relationships (pre-registered as H3). Paralleling findings for experience, in communal relationships, participants reported expressing putatively engaging emotions more frequently than disengaging ones, but in transactional relationships, they reported expressing the emotion types at similar frequencies (pre-registered as H4). All four emotions were expressed more frequently in communal relationships. The difference in reported expression between communal and transactional relationships was significantly larger for respect and pride than for guilt and anger. These results are presented in Figure 1 and Table 2.

Participants reported feeling more connected when expressing emotions in communal versus transactional relationships ($\beta = 0.54$, $t(4561) = 16.54$, $p < .001$, 95% CI $[0.473, 0.600]$, $d = 0.495$; pre-registered as H5) and when expressing putatively engaging versus disengaging emotions ($\beta = 0.68$, $t(4561) = 21.45$, $p < .001$, 95% CI $[0.614, 0.737]$, $d = 0.635$; pre-registered as H6). For all four emotions, participants reported feeling more connected when expressing them in communal than in transactional relationships. The difference in reported feelings of

connection between relationship types was significantly higher for respect than for pride, and this difference was higher for pride than for anger and guilt. Additionally, expressing guilt was reported as connecting in communal relationships but disconnecting in transactional contexts. That is, the reported feelings of connection when expressing guilt to communal partners was significantly above zero (i.e., associated with feelings of connection; $M = 0.21$, $t(498) = 4.66$, $p < .001$, 95% CI [0.119, 0.293], $d = 0.21$), but the reported feelings of connection when expressing guilt to transactional partners were significantly lower than zero (i.e., associated with feelings of disconnection; $M = -0.17$, $t(268) = -3.03$, $p = .003$, 95% CI [-0.282, -0.060], $d = 0.18$). These results are presented in Figure 2, see also Table 3.

Finally, the interaction between reported feelings of connection and relationship type predicting reported frequency of expression was significant ($\beta = -0.14$, $t(4553) = -6.08$, $p < .001$, 95% CI [-0.190, -0.097]; pre-registered as H7). Specifically, for each unit increase in reported feelings of connection when expressing an emotion, the increase in reported frequency of expressing that emotion was larger in communal ($\beta = 0.52$, $t(4553) = 36.40$, $p < .001$, 95% CI [0.495, 0.551]) than transactional relationships ($\beta = 0.38$, $t(4553) = 20.25$, $p < .001$, 95% CI [0.343, 0.417]), although this increase was still significant for transactional relationships, unlike predicted. After controlling for reported feelings of connection and its interaction with relationship type, there was no credible evidence for a difference between communal and transactional relationships in reported frequency of expression ($\beta = -0.01$, $t(4553) = -0.35$, $p = .73$, 95% CI [-0.061, 0.043]).

Study 2

Method

Hypotheses

There were six primary hypotheses for Study 2, largely informed by the findings from Study 1. The first three pertain to the reported frequency of perceived experience and expression; the latter three to the reported feelings of connection associated with perception.

Our first five hypotheses were based on the analogous results from Study 1. We hypothesized that participants would report perceiving emotion expression and experience more frequently in communal (versus transactional) relationships (H1; H2). Additionally, we hypothesized that the difference between reported frequency of perceived expression and experience would be higher for transactional (versus communal) relationships and for putatively disengaging (versus engaging) emotions (H3). Next, we hypothesized that participants would report feeling more connected to their communal (versus transactional) partners when perceiving them express emotion (H4) and that participants would report feeling more connected with their relational partners when perceiving them express putatively socially engaging (versus disengaging) emotions (H5).

The last hypothesis involves a comparison between Study 1 and Study 2. Specifically, we hypothesized that participants would report feeling more connected to their relationship partners when expressing an emotion to them (from Study 1), compared to when perceiving the same partner express the same emotion (H6). Our rationale for this hypothesis is our belief that individuals generally choose to express emotions to others for a reason (like facilitating connection), but they cannot fully control the emotions that are expressed to them.

Participants

The recruitment procedure, eligibility requirements, and recruitment sample size were the same for Study 2 as in Study 1, except for the additional requirement that individuals were only eligible to participate in Study 2 if they did not take part in Study 1.

After removing data from participants who failed at least one attention check ($n = 9$), we retained responses from 394 unique individuals. 67.5% were White, 10.2% were multiple races, 10.2% were Black, 6.3% were Hispanic, 4.3% were Asian, 0.5% were Middle Eastern, 0.5% were some other race, 0.0025% were American Indian, and 0.0025% preferred not to disclose their race. The mean age of retained participants was 38.2 years ($SD = 11.1$). 66.8% were women, 31.0% were men, 1.8% were non-binary, 0.025% were some other gender, and 0.025% preferred not to report their sex.

Procedure

Consent to participate and all measures for Study 2 were administered via an online anonymous Qualtrics survey. Participants first read short descriptions about experiencing and expressing emotions. These descriptions were nearly identical to those that participants in Study 1 read, except the focus was now on other people experiencing and expressing emotions. Then, participants responded to the same sixteen emotion-relationship scenarios as in Study 1. However, the emotions were modified so that the relationship partner was experiencing the emotion instead of the participant. The emotions were “respect for you”, “guilt toward you”, “pride in themselves”, and “anger at you”. As mentioned in the introduction, Kitayama et al. (2006) classified the former two emotions as engaging and the latter two as disengaging.

Question order randomization, data scoring, and attention check procedure was the same in Study 2 as in Study 1. However, the order of the experience and expression questions were

reversed in Study 2 compared to Study 1. That is, participants were first asked to report how frequently their relationship partners express each emotion when they are with the participant, then they were asked to report how frequently their relationship partners experience the given emotion when they are with the participant. Individuals generally infer others to be experiencing emotions after they perceive them as expressing an emotion (Lange et al., 2022). This was done so the ordering of the questions mirrored the typical emotion inference process.

Results

Participants reported their communal partners express ($\beta = 0.53$, $t(6300) = 18.26$, $p < .001$, 95% CI [0.476, 0.589], $d = 0.466$; pre-registered as H1), and experience ($\beta = 0.52$, $t(6298) = 17.94$, $p < .001$, 95% CI [0.465, 0.577], $d = 0.460$; pre-registered as H2) emotions more frequently than their transactional ones.

There was no significant difference between the reported frequencies of partner emotion expression and experience ($\beta = 0.01$, $t(12600) = 0.42$, $p = .676$, 95% CI [-0.032, 0.049], $d = 0.007$). There was also no significant interaction between relationship type and reported frequency of partner expression/experience ($\beta = -0.01$, $t(12598) = -0.294$, $p = .769$, 95% CI [-0.091, 0.067]) and the interaction between emotion type and reported frequency of partner expression/experience was only marginally significant ($\beta = 0.08$, $t(12598) = 1.88$, $p = .06$, 95% CI [-0.003, 0.159]; inconsistent with our pre-registered H3). For pairwise comparisons and statistics, see the online supplemental materials.

Participants reported feeling more connected when perceiving communal (versus transactional) partners express emotion ($\beta = 0.62$, $t(5068) = 19.63$, $p < .001$, 95% CI [0.554, 0.677], $d = 0.550$; pre-registered as H4) and when perceiving putatively engaging (versus disengaging) emotions ($\beta = 0.71$, $t(5068) = 23.18$, $p < .001$, 95% CI [0.654, 0.775], $d = 0.65$;

pre-registered as H5). However, in comparing Studies 1 and 2, participants actually reported feeling less connected with their relationship partners when expressing emotions compared to when perceiving their partners express emotions ($\beta = -0.05$, $t(9631) = -2.30$, $p = .02$, 95% CI [0.009, 0.099], $d = 0.05$; inconsistent with pre-registered H6).

Again, we assessed differences between relationship types for each emotion. Participants reported that communal partners experience all four emotions more frequently than transactional ones. The difference in frequencies of partner experience between relationship types was significantly higher for respect than for the other three emotions. These results are presented in Figure 1 and Table 4. Participants also reported that communal partners express all four emotions more frequently than transactional partners. The difference in frequencies of partner expression between relationship types was significantly higher for respect than for pride and guilt, and for guilt than for anger. These results are presented in Figure 1 and Table 5. Finally, perceptions of all four emotions were associated with greater feelings of connection in communal (versus transactional) relationships. The difference between relationship types in reported feelings of connection when perceiving was the largest for pride and the smallest for guilt. These results are presented in Figure 2 and Table 6.

Study 3

Method

Hypotheses

We tested six primary hypotheses in Study 3 that were largely informed by the results and rationale from Studies 1 and 2. We hypothesized participants would report experiencing and expressing the new set of emotions (empathic joy, empathic sadness, hurt, love) more frequently in communal than in transactional relationships (H1, H2). We also hypothesized that participants

would report feeling more connected when expressing these emotions to communal (versus transactional) partners (H3). Similarly, we hypothesized that participants would report these emotions as relationship-affirming in communal relationships but relationship-undermining in transactional ones (H4) and that expressing these emotions draws them closer to communal partners but distances them from their transactional ones (H5).

We also tested four secondary hypotheses in Study 3.⁴ We hypothesized that participants would report being more willing to express these emotions to communal (versus transactional) partners (EH1). In addition, we hypothesized that individuals who report never having expressed a given emotion to a given partner will report that emotion as more disconnecting (EH2), distancing (EH3), and relationship-undermining (EH4) than participants who have expressed that emotion in that context. Our rationale is that those who choose not to express a given emotion to a given partner will often do so because they believe expressing it would harm the relationship.

Participants

We conducted a similar power analysis for Study 3 as we did with Study 1, determining that a sample size of 368 would sufficiently power us to detect an effect of relationship type for each interpersonal measure for each emotion. The recruitment procedure and eligibility requirements were the same for Study 3 as in prior studies, except for the additional requirement that individuals were only eligible to participate in Study 3 if they did not take part in Studies 1 or 2.

After removing data from participants who failed at least one attention check ($n = 15$), we retained responses from 353 unique individuals. 69.7% were White, 11.9% were Black, 9.6% were multiple races, 4.2% were Asian, 4.0% were Latino, 2.3% were some other ethnicity, 0.3%

⁴ Pre-registered as exploratory hypotheses

were Middle Eastern, and 0.3% did not disclose their race. The mean age of retained participants was 39.93 years ($SD = 12.6$). 67.1% of retained participants were women, 28.6% were men, and 4.2% were non-binary.

Procedure

The procedure for Study 3 was largely the same as for Study 1, except the emotions of interest were now the four emotions chosen for their centrality to the care function of communal relationships (hurt, empathic joy, empathic sadness, and love). Specifically, empathic joy and empathic sadness were chosen because both indicate that the person who feels and expresses these emotions wishes to understand and care for a partner's successes (joy) and losses (sadness; Morelli et al., 2015). Hurt was chosen because it indicates that a partner expected more care from the partner than was received, and they want the partner to redress that situation (Lemay et al., 2012). Lastly, love was chosen because it is often elicited by caring for and being cared by another (Graham, 1983).

New to this study were measures related to the functional nature of emotion expression and willingness to express emotion across relational contexts. As before, individuals reported on how connected they feel when expressing each emotion to each partner, but they also reported on the extent to which each emotion expression affirms/undermines their relationship with their partner and the extent to which each expression draws them close versus distances them from their partner. Whereas the former has connotations related to the care function specifically, the second, we believe, more broadly assesses whether a given emotion is functional for any type of relationship. This provides a relatively strong test of the hypothesis that the emotions of empathic joy, empathic sadness, hurt, and love are particularly functional for communal relationships because this question could pick up on other potential relational functions of those

emotions. When individuals indicated that they “never” express a given emotion to a given partner, they responded to modified versions of these three measures asking participants to report what these interpersonal effects would be if they did express these emotions in these contexts. These measures were presented in random order for each emotion-relationship combination. Lastly, for each emotion-relationship block, participants reported on how willing they are to express each emotion to each partner.

Results

Participants reported experiencing ($\beta = 1.27$, $t(5641) = 43.2$, $p < .001$, 95% CI [1.22, 1.33], $d = 1.15$; pre-registered as H1) and expressing ($\beta = 1.44$, $t(5643) = 48.1$, $p < .001$, 95% CI [1.38, 1.50], $d = 1.28$; pre-registered as H2) emotions more frequently in communal than transactional relationships. For pairwise comparisons, see the online supplemental materials.

Participants reported emotions were more connecting ($\beta = 0.94$, $t(5643) = 39.28$, $p < .001$, 95% CI [0.898, 0.992], $d = 1.05$; pre-registered as hypothesis 3), more relationship-affirming ($\beta = 1.05$, $t(5631) = 43.60$, $p < .001$, 95% CI [1.00, 1.09], $d = 1.16$), and drew them closer ($\beta = 0.93$, $t(5639) = 37.73$, $p < .001$, 95% CI [0.89, 0.98], $d = 1.00$) in communal than transactional relationships. However, participants reported these emotions as relationship-affirming (significantly above zero) in both communal ($M = 1.25$, $t(2816) = 77.69$, $p < .001$, 95% CI [1.22, 1.28], $d = 1.46$; pre-registered as H4) and transactional relationships ($M = 0.20$, $t(2815) = 11.35$, $p < .001$, 95% CI [0.17, 0.24], $d = 0.21$; inconsistent with H4), but to a much smaller degree in transactional ones. Similarly, participants reported these emotions draw them closer (significantly above zero) in both communal ($M = 1.21$, $t(2822) = 70.98$, $p < .001$, 95% CI [1.17, 1.24], $d = 1.34$; pre-registered as H5) and transactional relationships ($M = 0.27$, $t(2817) = 15.09$, $p < .001$, 95% CI [0.24, 0.31], $d = 0.28$; inconsistent with H5), but to a much smaller

degree in transactional ones. Individuals were also more willing to express emotion to communal than transactional partners ($\beta = 1.31$, $t(5638) = 44.86$, $p < 0.001$, 95% CI [1.25, 1.38], $d = 1.02$; pre-registered as EH1). These results are presented in Figure 3.

Participants who report never expressing a given emotion to a given partner believed that doing so would be more disconnecting ($\beta = -1.22$, $t(5637) = -39.65$, $p < .001$, 95% CI [-1.28, -1.16], $d = 1.36$; pre-registered as EH2), more relationship-undermining ($\beta = -1.30$, $t(5631) = -41.80$, $p < .001$, 95% CI [-1.24, -1.36], $d = 1.43$; pre-registered as EH4) and more distancing ($\beta = -1.27$, $t(5639) = -40.61$, $p < .001$, 95% CI [-1.33, -1.21], $d = 1.39$; pre-registered as EH3) than participants who report expressing that emotion to that partner recall it is. Comparisons with zero revealed that participants who report never expressing a given emotion to a given partner believed that this expression would be disconnecting ($M = -0.24$, $t(1048) = -7.72$, $p < .001$, 95% CI [-0.30, -0.18], $d = 0.24$), relationship-undermining ($M = -0.33$, $t(1049) = -11.19$, $p < .001$, 95% CI [-0.39, -0.27], $d = 0.34$), and distancing ($M = -0.30$, $t(1047) = -9.50$, $p < .001$, 95% CI [-0.35, -0.25], $d = 0.29$), while participants who report expressing a given emotion to a given partner recalled doing so as connecting ($M = 0.99$, $t(1048) = 76.04$, $p < .001$, 95% CI [0.96, 1.01], $d = 1.12$), relationship-affirming ($M = 0.97$, $t(4582) = 73.07$, $p < .001$, 95% CI [0.94, 0.995], $d = 1.08$), and as drawing them closer to their partner ($M = 0.98$, $t(4592) = 74.13$, $p < .001$, 95% CI [0.95, 1.00], $d = 1.09$).

Since the three measures pertaining to the functional nature of reported emotion expression depended on relationship type and the recalled/hypothetical nature of the reported expression, we conducted moderation analyses to see if the recalled/hypothetical nature of reported expression moderates the effect of relationship type on each measure of emotion function. A logistic regression analysis showed that the odds of participants reporting they never

express a given emotion were approximately 11.6 times higher (odds ratio = 0.086, 95% CI [0.070, 0.104]) in transactional than communal relationships ($\beta = -2.45$, $z = -23.77$, $p < .001$, 95% CI [-2.66, -2.26]). Lastly, whether participants report having expressed a given emotion to a given partner significantly moderated the effect of relationship type on the extent to which the emotion was connecting ($\beta = -0.53$, $t(5635) = -6.122$, $p < .001$, 95% CI [-0.70, -0.36]), was relationship-affirming ($\beta = -0.49$, $t(5635) = -5.716$, $p < .001$, 95% CI [-0.66, -0.32]), and draws participants closer ($\beta = -0.66$, $t(5637) = -7.461$, $p < .001$, 95% CI [-0.84, -0.49]).

Study 4

Method

Hypotheses

We tested six primary hypotheses in Study 4 that were largely analogous to the ones from Study 3, except we no longer made predictions about the group difference of each dependent variable in comparison to zero.

We hypothesized that participants would report perceiving emotion expression and experience more frequently in communal (versus transactional) relationships (H1, H2). We also hypothesized that participants would report perceiving these emotions as more connecting (H3), relationship-affirming (H4), and as drawing them closer (H5) in communal (versus transactional) relationships.

There were four exploratory hypotheses tested in this study that were analogous to the exploratory hypotheses from Study 3. We hypothesized that participants would report their communal partners as more willing to express these emotions than their transactional ones (EH1). We also hypothesized that individuals who report never perceiving a given emotion expressed by a given partner would believe that perception in that context would be less

connecting (EH2), more distancing (EH3), and more relationship-undermining (EH4) than participants who have perceived that emotion expressed by that partner recall it is.

Participants

The recruitment procedure, eligibility requirements, and recruitment sample size were the same for Study 4 as in Study 3, except for the additional requirement that individuals were only eligible to participate in Study 4 if they did not take part in Studies 1, 2, or 3.

After removing data from participants who did not finish the survey ($n = 1$) and participants who failed at least one or more attention checks ($n = 37$), we retained data from 330 unique individuals. 75.5% were White, 7.9% were Black, 7.2% were multiple races, 3.9% were Latino, 3.0% were Asian, 1% were American Indian, 0.6% were Middle Eastern, 0.6% were some other ethnicity, and 0.3% (1 participant) did not disclose their race. The mean age of retained participants was 40.06 years ($SD = 12.4$). 65.8% of retained participants were women, 30.9% were men, 3.0% were non-binary, and 0.3% were some other gender.

Procedure

The procedure for Study 4 was the same as for Study 3, except the emotions of interest were modified so that the relationship partner was experiencing the emotion. That is, in the context of the same four relationship partners as in the previous studies, individuals were asked how frequently their partners express and experience hurt to them, empathic joy for them, empathic sadness for them, and love for them. As in Study 2, perceived expression questions were asked before perceived experience questions.

Results

Participants reported their communal partners express ($\beta = 1.53$, $t(5272) = 53.17$, $p < .001$, 95% CI [1.47, 1.59], $d = 1.46$; pre-registered as H1) and experience ($\beta = 1.40$, $t(5268) =$

48.8, $p < .001$, 95% CI [1.34, 1.46], $d = 1.34$; pre-registered as H1) emotions more frequently than their transactional ones. For pairwise comparisons, see the online supplemental materials.

Participants reported that communal partners expressing emotion was more connecting ($\beta = 1.03$, $t(5274) = 44.74$, $p < .001$, 95% CI [0.987, 1.08], $d = 1.23$; pre-registered as H3), more relationship-affirming ($\beta = 1.16$, $t(5274) = 51.21$, $p < .001$, 95% CI [1.12, 1.21], $d = 1.41$; pre-registered as H4), and drew individuals closer ($\beta = 1.03$, $t(5274) = 43.54$, $p < .001$, 95% CI [0.99, 1.08], $d = 1.20$; pre-registered as H5) than when transactional partners express emotion.

Participants also reported their communal partners as more willing to express these emotions than their transactional ones ($\beta = 1.25$, $t(5269) = 47.80$, $p < .001$, 95% CI [1.20, 1.31], $d = 1.32$; pre-registered as EH1). These results are presented in Figure 4.

Participants who report never perceiving a given emotion expressed by a given partner believed this emotion in this context would be more disconnecting ($\beta = -1.24$, $t(5274) = -39.58$, $p < 0.001$, 95% CI [-1.30, -1.18], $d = 1.43$; pre-registered as EH2), relationship-undermining ($\beta = -1.25$, $t(5274) = -38.75$, $p < 0.001$, 95% CI [-1.32, -1.19], $d = 1.40$; pre-registered as EH4), and distancing ($M = -0.22$, $t(922) = -6.41$, $p < 0.001$, 95% CI [-0.29, -0.15], $d = 0.21$; pre-registered as EH3) than individuals who report having perceived this emotion in this context recall it is.

Comparisons with zero revealed that participants who report a given partner never having expressed a given emotion to them believed doing so would be disconnecting ($M = -0.17$, $t(922) = -5.17$, $p < .001$, 95% CI [-0.24, -0.11], $d = 0.17$), relationship-undermining ($M = -0.25$, $t(922) = -7.79$, $p < .001$, 95% CI [-0.31, -0.19], $d = 0.26$), and distancing ($M = -0.22$, $t(922) = -6.41$, $p < 0.001$, 95% CI [-0.29, -0.15], $d = 0.21$), while participants who report having perceived a given emotion expressed by a given partner recalled it as connecting ($M = 1.07$, $t(4352) = 85.07$, $p < .001$, 95% CI [1.04, 1.09], $d = 1.29$), relationship-affirming ($M = 1.0$, $t(4352) = 75.72$, $p < .001$,

95% CI [0.98, 1.03], $d = 1.15$), and as drawing them closer ($M = 1.03$, $t(4356) = 79.58$, $p < .001$, 95% CI [1.00, 1.05], $d = 1.20$).

Since the three measures pertaining to the functional nature of emotion perception depended on relationship type and the recalled/hypothetical nature of the perception, we conducted moderation analyses to see if the recalled/hypothetical nature of reported perception moderates the effect of relationship type on each measure of emotion function. A logistic regression analysis showed that the odds of participants reporting they never perceive a given emotion were approximately 15.38 times higher (odds ratio = 0.065, 95% CI [0.050, 0.082]) in transactional than communal relationships ($\beta = -2.74$, $z = -22.41$, $p < .001$, 95% CI [-2.99, -2.51]). Lastly, whether participants reporting having perceived a given emotion expressed by a given partner significantly moderated the effect of relationship type on the extent to which the emotion was connecting ($\beta = -0.53$, $t(5272) = -5.569$, $p < .001$, 95% CI [-0.72, -0.35]), was relationship-affirming ($\beta = -0.49$, $t(5272) = -5.132$, $p < .001$, 95% CI [-0.67, -0.30]), and draws participants closer ($\beta = -0.52$, $t(5276) = -5.314$, $p < .001$, 95% CI [-0.71, -0.33]).

General Discussion

In this paper, we aimed to demonstrate that the functions that relationships serve will drive the experience, expression, perception, and engaging or disengaging nature of eight emotions (respect, guilt, pride in self, anger, empathic joy, empathic sadness, hurt, and love). All four studies yielded evidence in support of our theoretical position that a range of experiences and expressions of emotion are more common and perceived to be more functional in communal than in transactional relationships. First, replicating existing work (e.g., Le & Reis, 2025; Von Culin et al., 2018; Ruan et al., 2020; Ruan et al., 2024), we find Americans report experiencing, expressing, and perceiving a range of emotions more frequently in communal than in

transactional relational contexts. We also find that Americans' reported feelings of connection when expressing and perceiving these emotions depend on relational context, such that emotions, generally, are more connecting in communal than transactional relationships. A striking example is guilt, which Americans reported as connecting when expressed within a communal relationship context but disconnecting when expressed within a transactional one. These findings support the claim that emotions are not inherently engaging or disengaging, but rather this aspect of emotion depends upon social contexts like relational context.

We then sought to understand what drives the higher frequency of emotions and their more engaging nature in communal than transactional contexts. We expected these tendencies to be driven by the extent to which emotion support the functions of communal relationships (i.e., care) more than those of transactional relationships. Thus, in Studies 3 and 4 we focused on a set of emotions with some of the strongest evidence of serving communal relationship functions, as we expected them to most clearly show contrasts across the relationship types. We find, once again, that Americans report these emotions as more connecting in communal than transactional relationships. Conceptually replicating this finding, we also find that these emotions draw individuals closer to the other in communal than in transactional relationships. We also broadened our initial focus based on feelings of connection to examine a broader measure of the relational function of emotions, finding that Americans report these same emotions are more relationship-affirming within communal versus transactional relational contexts. It seems that Americans recognize that the functional nature of the emotions depends on relational context. Consistent, Americans also report being much more willing to express these emotions in communal than transactional contexts and believe their communal partners to be much more willing to express these emotions to them than their transactional ones. Importantly, those who

never express or perceive these emotions in the context of a given partner reported that they believed these expressions and perceptions would be disconnecting, relationship-undermining, and distancing. Of course, the likelihood of reporting they never express or perceive these emotions was far higher in transactional than communal relationships. In turn, the functional nature of emotion provides a potential mechanism for why emotions are more frequent in communal than transactional relationships: Americans find emotions to be more functional in communal relationships than in transactional relationships, and they are more likely to express and perceive functional emotions.

Limitations and Future Directions

Some limitations of our work are the usage of retrospective self-report measures and the exclusive focus on Americans. Retrospective reports may be informed more so by beliefs about emotions than on actual instances of these emotions (Robinson & Clore, 2002), so the reports from the current work may diverge from momentary beliefs about the functional natures of these emotions across relationship types. As such, it is possible that the findings reported here reflect what people believe about emotions and relationships generally, but not necessarily the reactions to these emotions in the moment. Further, Oishi (2002) showed that European Americans (the majority of our four samples) are especially likely to have a positivity bias in retrospective reports regarding well-being, so it is possible that Americans in our samples overreported the functional nature of the emotions in the present research. However, given that emotions were reported as functional in communal relationships but often dysfunctional in transactional ones, a positivity bias in our participants' reports only seems plausible when considering communal contexts. This, in and of itself, has interesting implications for how Americans think about the engaging/disengaging nature of emotions across relational contexts that largely matches our

functional account. Future work should employ methodologies such as ecological momentary assessment across cultures to replicate our findings and show that relationship context shapes the functional and engaging nature of emotions both retrospectively and in the moment.

Recent work has examined emotions across both cultures and relationships, but this work has largely not included measures of the functions of emotions across social contexts. Salvador et al. (2024) examined how cross-cultural interdependence styles relate to emotions in interpersonal versus personal situations. They found that both European and Latin Americans expressed more putatively disengaging emotions when a positive event happened to them alone, and that this tendency was more pronounced for European than Latin Americans. However, when a negative event happened to a close other, individuals in both cultural groups expressed more putatively engaging emotions, and this tendency was more pronounced for Latin than European Americans. Boiger et al. (2022) found that Belgian couples who experienced a high number of putatively disengaging emotions during disagreements (aligning with the cultural norm) reported higher levels of autonomy and relationship satisfaction. Sunar et al. (2021) have further shown that violations of different relational norms produce different intensities of “moral emotions,” and this holds true among both Turkish and English-speaking samples. So, there is strong evidence that emotion patterning depends both upon the culture’s dominant interdependence pattern and relational context. Future work combining the present approach with such cross-cultural tests could examine whether social functions that emotions serve across cultures and relationships explain these patterns.

Similarly, whether a relational partner is also experiencing the same emotion could impact the self-reported interpersonal functions of an emotion’s expression and perception. We expect this to be especially true when partners are responding to the same situation (i.e., are

experiencing a collective emotion; Goldenberg et al., 2020). Future work should explore this possibility.

Conclusions

Americans report more frequently experiencing, expressing, and perceiving emotions in communal than in transactional relationship contexts, and this seems to be driven by emotions being perceived as more functional and engaging in communal contexts. That is, since the form of interdependence defining communal relationships is one of mutual care and responsiveness, emotions often facilitate the functions of these relationships while often obstructing the functions of others (like transactional ones). Therefore, the extent to which an emotion is engaging or disengaging depends upon the functions of the relationship within which it is experienced, expressed, and perceived, such that emotions, generally, are more engaging in communal relationships.

Declarations

Ethics and consent to participate

All participants consented to participate in the presented studies, and all studies were approved by Yale University's IRB, approval # 2000028669.

Consent for publication

The presented results have no identifying details that prevent it from it from publication.

Availability of data and material

Deidentified data and all materials are available on the Open Science Framework at <https://osf.io/jcxat/>.

Competing interests

The authors have no competing interests.

Funding

The authors have no external funding to declare

Author contributions

MKC drafted the manuscript and created the figures. MSC and MG provided critical revisions, and all three authors planned the studies and approved the final manuscript for submission.

Acknowledgements

Not applicable.

References

- Barrett, L. F., Robin, L., Pietromonaco, P. R., & Eyssell, K. M. (1998). Are women the "more emotional" sex? Evidence from emotional experiences in social context. *Cognition and Emotion*, 12(4), 555–578. <https://doi.org/10.1080/026999398379565>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Boiger, M., Kirchner-Häusler, A., Schouten, A., Uchida, Y., & Mesquita, B. (2022). Different bumps in the road: The emotional dynamics of couple disagreements in Belgium and Japan. *Emotion*, 22(5), 805–819. <https://doi.org/10.1037/emo0000910>
- Clark, M. S. (1984). Record keeping in two types of relationships. *Journal of Personality and Social Psychology*, 47(3), 549–557. <https://doi.org/10.1037/0022-3514.47.3.549>
- Clark, M. S., Adkins, C., Chughtai, M. K., & Elizabeth, H. (2025). How communal relational contexts shape (and are shaped by) emotional lives. In A. L. Vangelisti, D. Perlman, & S. Sprecher (Eds.), *Cambridge handbook of personal relationships*. Cambridge University Press.
- Clark, M. S., & Mills, J. (1979). Interpersonal attraction in exchange and communal relationships. *Journal of Personality and Social Psychology*, 37(1), 12–24. <https://doi.org/10.1037/0022-3514.37.1.12>
- Clark, M. S., & Mills, J. R. (2012). A theory of communal (and exchange) relationships. In P. A. M. Van Lange, A. W. Kruglanski, & E. T. Higgins (Eds.), *Handbook of theories of social psychology* (pp. 232–250). Sage Publications Ltd. <https://doi.org/10.4135/9781446249222.n38>

- Clark, M. S., Mills, J. R., & Corcoran, D. M. (1989). Keeping track of needs and inputs of friends and strangers. *Personality and Social Psychology Bulletin*, 15(4), 533–542.
<https://doi.org/10.1177/0146167289154007>
- Clark, M. S., Ouellette, R., Powell, M. C., & Milberg, S. (1987). Recipient's mood, relationship type, and helping. *Journal of Personality and Social Psychology*, 53(1), 94–103.
<https://doi.org/10.1037//0022-3514.53.1.94>
- Clark, M. S., & Taraban, C. (1991). Reactions to and willingness to express emotion in communal and exchange relationships. *Journal of Experimental Social Psychology*, 27(4), 324–336. [https://doi.org/10.1016/0022-1031\(91\)90029-6](https://doi.org/10.1016/0022-1031(91)90029-6)
- Earp, B. D., McLoughlin, K. L., Monrad, J. T., et al. (2021). How social relationships shape moral wrongness judgments. *Nature Communications*, 12(1), 5776.
<https://doi.org/10.1038/s41467-021-26067-4>
- Earp, B.D., McLoughlin, K.L., Calcott, R., Caraccio, M., Monrad, J.T., Owen, A.M., Crockett, M.J. & Clark, M.S. (2025, submitted). How social relationships shape Praise and blame: Strengthening and extending the relational norms model.
- Fiske, A. P. (1992). The four elementary forms of sociality: Framework for a unified theory of social relations. *Psychological Review*, 99(4), 689–723. <https://doi.org/10.1037/0033-295X.99.4.689>
- Goldenberg, A., Garcia, D., Halperin, E., & Gross, J. J. (2020). Collective emotions. *Current Directions in Psychological Science*, 29(2), 154–160.
<https://doi.org/10.1177/0963721420901574>
- Graham, H. (1983). *Caring: A labour of love*. In *A labour of love* (1st ed., pp. 18). Routledge.
<https://doi.org/10.4324/9781003303930>

- Kitayama, S., Mesquita, B., & Karasawa, M. (2006). Cultural affordances and emotional experience: Socially engaging and disengaging emotions in Japan and the United States. *Journal of Personality and Social Psychology*, 91(5), 890–903.
<https://doi.org/10.1037/0022-3514.91.5.890>
- Lange, J., Heerdink, M. W., & van Kleef, G. A. (2022). Reading emotions, reading people: Emotion perception and inferences drawn from perceived emotions. *Current Opinion in Psychology*, 43, 85–90. <https://doi.org/10.1016/j.copsyc.2021.06.008>
- Le, J. D. V., & Reis, H. T. (2024). Supporting the willingness to express emotions in relationships: The role of perceived empathic effort and interpersonal accuracy. *Emotion*, 24(5), 1338–1342. <https://doi.org/10.1037/emo0001354>
- Lemay, E. P., Jr., Overall, N. C., & Clark, M. S. (2012). Experiences and interpersonal consequences of hurt feelings and anger. *Journal of Personality and Social Psychology*, 103(6), 982–1006. <https://doi.org/10.1037/a0030064>
- Lüdecke, D. (2024). sjstats: Statistical functions for regression models (Version 0.19.0). <https://CRAN.R-project.org/package=sjstats>
- Morelli, S. A., Lieberman, M. D., & Zaki, J. (2015). The emerging study of positive empathy. *Social and Personality Psychology Compass*, 9(2), 57–68. <https://doi.org/10.1111/spc3.12157>
- Oishi, S. (2002). The experiencing and remembering of well-being: A cross-cultural analysis. *Personality and Social Psychology Bulletin*, 28(10), 1398–1406.
<https://doi.org/10.1177/014616702236871>

- R Core Team. (2024). R: A language and environment for statistical computing (Version 4.4.1) [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>
- R Core Team. (2024). stats: The R stats package (Version 4.4.1) [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Rai, T. S., & Fiske, A. P. (2011). Moral psychology is relationship regulation: Moral motives for unity, hierarchy, equality, and proportionality. *Psychological Review*, 118(1), 57–75. <https://doi.org/10.1037/a0021867>
- Robinson, M. D., & Clore, G. L. (2002). Belief and feeling: Evidence for an accessibility model of emotional self-report. *Psychological Bulletin*, 128(6), 934–960. <https://doi.org/10.1037/0033-2909.128.6.934>
- Ruan, Y., Le, J. D. V., & Reis, H. T. (2024). How can I help?: Specific strategies used in interpersonal emotion regulation in a relationship context. *Emotion*, 24(2), 329–344. <https://doi.org/10.1037/emo0001272>
- Ruan, Y., Reis, H. T., Clark, M. S., Hirsch, J. L., & Bink, B. D. (2020). Can I tell you how I feel? Perceived partner responsiveness encourages emotional expression. *Emotion*, 20(3), 329–342. <https://doi.org/10.1037/emo0000650>
- Rusbult, C. E., & Van Lange, P. A. (2003). Interdependence, interaction, and relationships. *Annual review of psychology*, 54, 351–375. <https://doi.org/10.1146/annurev.psych.54.101601.145059>
- Salvador, C. E., Idrovo Carlier, S., Ishii, K., Torres Castillo, C., Nanakdewa, K., San Martin, A., Savani, K., & Kitayama, S. (2024). Emotionally expressive interdependence in Latin

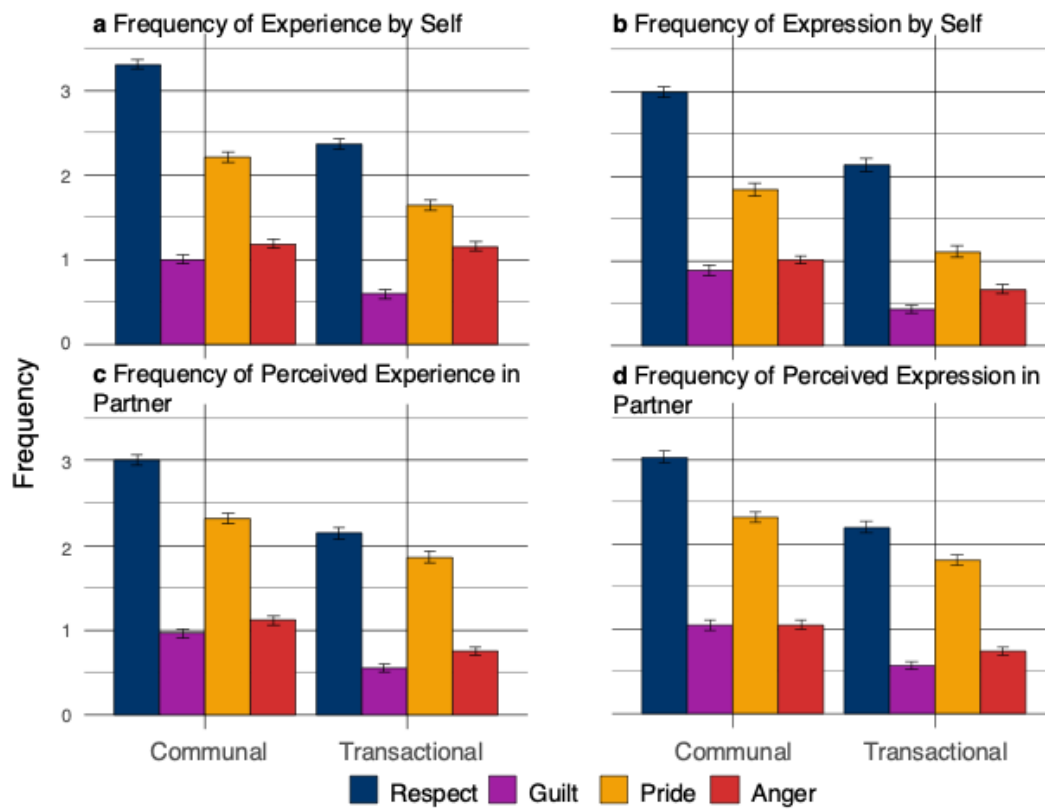
America: Triangulating through a comparison of three cultural zones. *Emotion*, 24(3), 820–835. <https://doi.org/10.1037/emo0001302>

Sunar, D., Cesur, S., Piyale, Z. E., Tepe, B., Biten, A. F., Hill, C. T., & Koç, Y. (2021). People respond with different moral emotions to violations in different relational models: A cross-cultural comparison. *Emotion*, 21(4), 693–706.
<https://doi.org/10.1037/emo0000736>

Von Culin, K. R., Hirsch, J. L., & Clark, M. S. (2018). Willingness to express emotion depends upon perceiving partner care. *Cognition and Emotion*, 32(3), 641–650.
<https://doi.org/10.1080/02699931.2017.1331906>

Figure 1

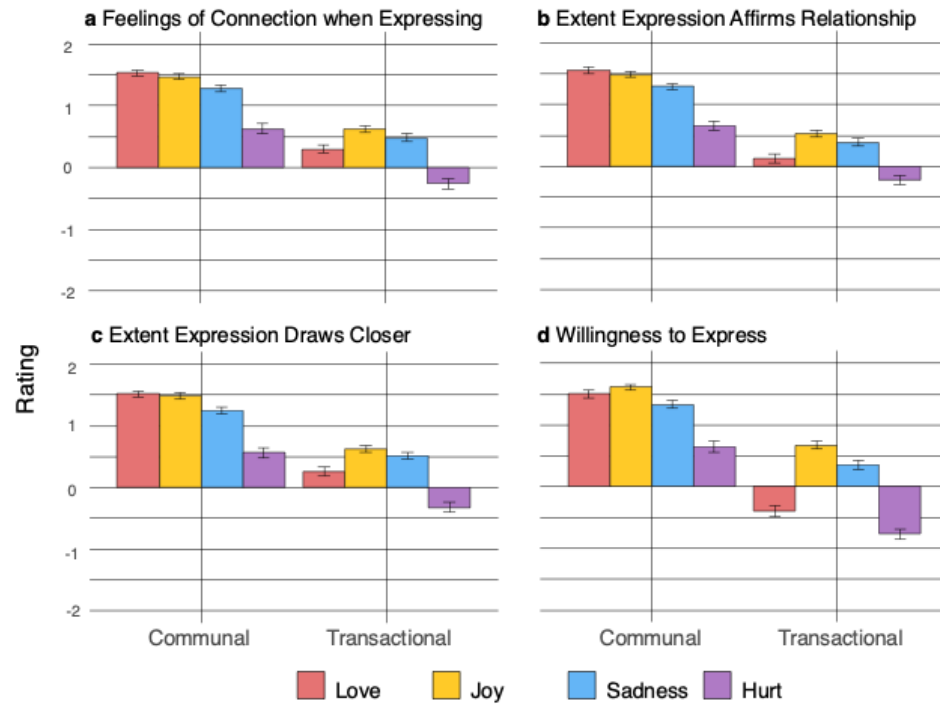
Study 1 and 2 Reported Frequency of Emotion Experience and Expression



Mean reported frequency of experience and expression by self (Study 1) and reported frequency of perceived experience and expression in partner (Study 2) by emotion and relationship type. Error bars represent 95% confidence intervals.

Figure 2*Study 1 and 2 Feelings of Connection*

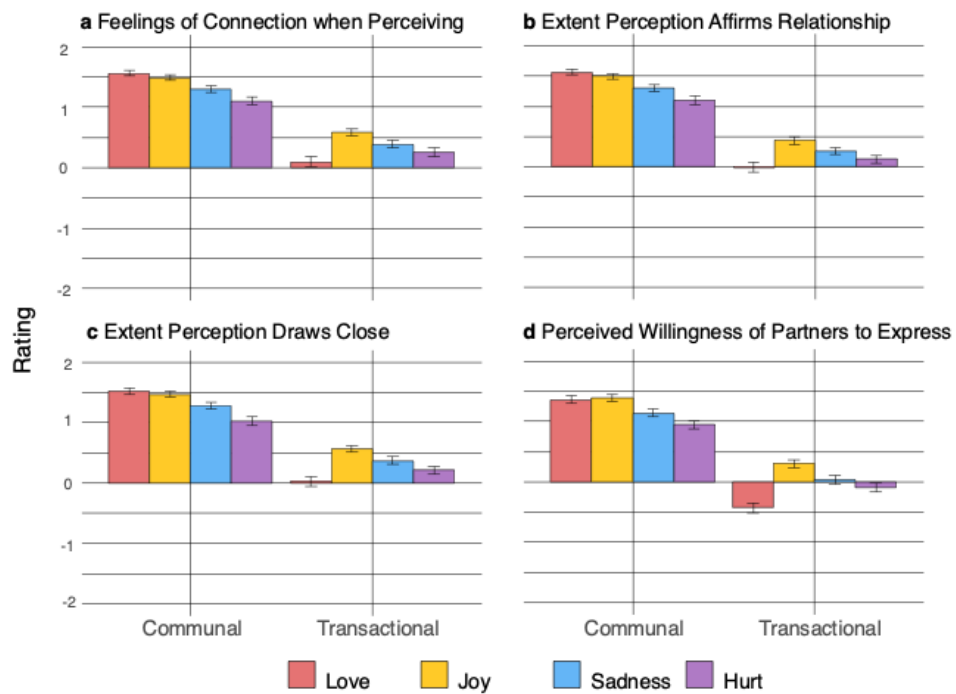
Mean reported feelings of connection when expressing (Study 1) and when perceiving (Study 2) by emotion and relationship type. Error bars represent 95% confidence intervals.

Figure 3*Study 3 Functional Nature of Expression and Willingness to Express*

Mean reported ratings of interpersonal functions of expression and willingness to express by emotion and relationship type (Study 3). Error bars represent 95% confidence intervals.

Figure 4

Study 4 Functional Nature of Partner Expression and Partner Willingness to Express



Mean reported ratings of interpersonal functions of partner expression and partner willingness to express by emotion and relationship type (Study 4). Error bars represent 95% confidence intervals.

Table 1

Study 1 post-hoc Tukey comparisons between relationship types and emotion types on reported frequencies of experiencing emotion types and individual emotions

Effect	β	<i>P</i>	95% CI	<i>d</i>
Engaging: Communal - Transactional	0.676	< .001	[0.571, 0.782]	0.586
Disengaging: Communal - Transactional	0.300	< .001	[0.195, 0.406]	0.260
Communal: Engaging - Disengaging	0.455	< .001	[0.349, 0.561]	0.395
Transactional: Engaging - Disengaging	0.079	.2173	[0.027, 0.185]	0.069
Respect: Communal - Transactional	0.942	< .001	[0.812, 1.08]	1.10
Pride: Communal - Transactional	0.565	< .001	[0.435, 0.696]	0.664
Guilt: Communal - Transactional	0.412	< .001	[0.282, 0.543]	0.484
Anger: Communal - Transactional	0.035	.993	[-0.096, 0.165]	0.041

Table 2

Study 1 post-hoc Tukey comparisons between relationship types and emotion types on reported frequencies of expressing emotion types and individual emotions

Effect	β	p	95% CI	d
Engaging: Communal - Transactional	0.662	< .001	[0.557, 0.766]	0.582
Disengaging: Communal - Transactional	0.538	< .001	[0.434, 0.642]	0.473
Communal: Engaging - Disengaging	0.516	< .001	[0.412, 0.621]	0.454
Transactional: Engaging - Disengaging	0.392	< .001	[0.288, 0.497]	0.345
Respect: Communal - Transactional	0.862	< .001	[0.727, 0.997]	0.977
Pride Communal - Transactional	0.731	< .001	[0.596, 0.866]	0.829
Guilt Communal - Transactional	0.457	< .001	[0.322, 0.592]	0.518
Anger Communal - Transactional	0.345	< .001	[0.210, 0.480]	0.391

Table 3

Study 1 post-hoc Tukey comparisons between relationship types and emotion types on reported feelings of connection when expressing emotion types and individual emotions

Effect	β	p	95% CI	d
Communal: Engaging - Disengaging	0.698	< .001	[0.615, 0.782]	0.644
Transactional: Engaging - Disengaging	0.660	< .001	[0.576, 0.745]	0.693
Engaging: Communal - Transactional	0.561	< .001	[0.481, 0.640]	0.587
Disengaging: Communal - Transactional	0.523	< .001	[0.431, 0.614]	0.475
Respect: Communal - Transactional	0.834	< .001	[0.701, 0.966]	1.09
Pride: Communal - Transactional	0.693	< .001	[0.547, 0.838]	0.817
Guilt: Communal - Transactional	0.377	< 0.001	[0.183, 0.572]	0.445

Anger: Communal - Transactional	0.406	< .001	[0.242, 0.569]	0.479
------------------------------------	-------	--------	----------------	-------

Table 4

Study 2 post-hoc Tukey comparisons between relationship types on reported frequencies of perceiving experience of individual emotions

Effect	β	p	95% CI	d
Respect Communal - Transactional	0.858	< .001	[0.732, 0.984]	1.038
Pride Communal - Transactional	0.460	< .001	[0.334, 0.586]	0.556
Guilt Communal - Transactional	0.407	< .001	[0.281, 0.534]	0.493
Anger Communal - Transactional	0.362	< .001	[0.235, 0.488]	0.437

Table 5

Study 2 post-hoc Tukey comparisons between relationship types on reported frequencies of perceiving expression of individual emotions

Effect	β	p	95% CI	d
Respect Communal - Transactional	0.834	< .001	[0.706, 0.961]	0.997
Pride Communal - Transactional	0.500	< .001	[0.373, 0.628]	0.599
Guilt Communal - Transactional	0.481	< .001	[0.353, 0.609]	0.575
Anger Communal - Transactional	0.314	< .001	[0.187, 0.442]	0.376

Table 6

Study 2 post-hoc Tukey comparisons between relationship types on reported feelings of connection when perceiving individual emotions

Effect	β	p	95% CI	d
Respect	0.728	< .001	[0.593, 0.863]	0.836
Communal - Transactional				
Pride	0.801	< .001	[0.663, 0.938]	0.922
Communal - Transactional				
Guilt	0.416	< .001	[0.242, 0.589]	0.477
Communal - Transactional				
Anger	0.622	< .001	[0.460, 0.784]	0.714
Communal - Transactional				